

Prime Flight / RampVision — technical map: models, module weight, dependencies, duplicated logic

6 October 2026 · Valentyn Fedorov · from the architecture review (docs/arch_review), the consumption matrix and compute audits (docs/analysis: module_consumption, module_compute, module_unit_needs), the real-time measurements (rt_module_cost, rt_environment, rt_optimisation_plan) and the GM / tracker as-is reports. Live milliseconds: one camera at 8 fps on an RTX 5070 Ti with the card kept busy, 125 ms budget per frame; post seconds: clean single jobs on a 36-min event.

1 · Which models run, where, and what each costs

Stage	Models and measured cost
General Model (one job per video, two decodes)	Three fp16 ONNX YOLO heads on every frame: GM YOLOv8m @1088 (apron classes; 4.3 ms pure inference, 8.5 ms busy), chocks YOLOv8 @1280 (22.6 MB; 5.6 ms pure, +3.0 ms on top of GM), vehicle YOLOv8m @1088 (52 MB; 8.3 ms pure, +5.8 ms). All three 19.2 ms busy, 11.1 ms through TensorRT . Beside them: MobileSAM vit_t (40.7 MB) mask of the main aircraft every frame for its motion state; entity YOLOv5 @1280 (14.7 MB) every 8th frame up to 40 hits; camera cone/wing EfficientNet-B0 (16.3 MB) on every post-arrival frame with the main aircraft, on CPU through PIL; tail classifier (0.4 MB). The second run re-decodes the whole video to label obstacle / side_obstacle rows and the main aircraft. Production: 207 ms per frame (1.7x slower than recording).
Tracker (one job per video, pixels every frame)	Three DeepSORT instances (belt loader, workers, GSE) with one re-id net `scripted_cpkt.t7` (46 MB) — 8.3 ms per frame; instance masks at (re)initialisation: plane YOLO11s-seg (20.6 MB) and BL/GSE YOLO26s-seg (23.4 MB); FAST keypoints + PyrLK optical flow on ≤250 points per object; `estimate_sigma` noise estimate every 2 s (2.3 ms; 100 ms per frame on the master pin). Per class busy: airplane 4.9 ms, +person 7.7, +belt loader 20.7, all four 26.2 (belt loader path 9.6, aircraft 4.0, GSE 2.4). Production pin 36–49 ms per frame; v2 byte-identical.
Modules with own models (13 of 27)	safety-vests : Swin-T orientation (3 classes, 256) + Swin-T vest (2 classes, 256) + ResNet-18 presence gate (192) + HSV rules, every person every frame — 18.8 ms, 5.2 cores . handrails-on-gse : EfficientNet-B0 handrail (224) + Unet MIT-B0 (320×256) + DWPose-I ONNX (288×384) + EfficientNet hand classifier — 12.5 ms, 4.9 cores . HRNet-W48 COCO 256×192 (269 MB, one identical file) in hand-signals, steering, lead-marshaller, wing-walkers , flip test on (two backbone passes per call). pin-verification : SimCC ResNet-50 288×384. gse-chocks : RTMPose-I ONNX per (GSE × person), not batched + EfficientNet-B0 difference classifier. aircraft-chocks : EfficientNet-B0 difference classifier (other weights) + template matching per rear wheel. safety-zone : MobileSAM encoder per re-seeding vehicle + its own norfair truck tracker. conditioned-air : YOLO11s-seg @992 every 16th frame (`best28_full.onnx` never loaded). fod-walk : MoveNet Thunder ONNX per person, not batched. safety-handrails : the same EfficientNet-B0 handrail classifier every 8th frame. walk-arounds : YOLOv8m-pose 640 per person crop (pre-departure: every frame, not batched) + Depth Anything V2 ViT-B 518×924 + MobileSAM for the TDV build + a small CNN; own Python environments. hair-policy : two EfficientNet-B0 (Pyarmor build, WSL, CPU). steering also a tsai InceptionTime time-series classifier. Loaded by nobody: HRNet-W48 384×288 (255 MB, in all four HRNet checkpoints), `wand_detector.pt`, `worker_type_classifier.pkl`, `v4_probe_head.npz`.

2 · The heaviest modules (live own work per frame · CPU · RAM · post job on the whole event · batch ms per recorded frame)

Module	live ms/frame	cores	RAM GB	post job s	batch ms	why
safety-vests	18.8	5.2	1.7	370	28.2	two Swin-T + ResNet-18 on every person of every frame, whole session; decides at session end
handrails-on-gse	12.5	4.9	2.8	249	30.7	four networks per belt-loader visit (classifier, segmentation, pose, hand classifier); session end
wing-walkers	5.1	5.0	1.9	111	3.2	HRNet-W48 on all persons from departure start to +60 s; p95 85 ms, latency p95 1.4 s
gse-chocks	3.3	2.0	2.4	76	6.1	RTMPose per GSE × person, not batched; difference classifier per 10 s; three tracked classes; session end
lead-marshaller	2.5	3.4	2.4	59	5.7	HRNet-W48 on all workers while the plane approaches; full-frame CLAEH copy; 96-frame buffer
safety-handrails	2.4	4.6	1.3	46	8.4	reads and converts every window frame, uses 1 of 8
hand-signals · safety-zone · aircraft-chocks	0.3 · 0.3 · 0.5 (pass 1 only)	2.8 · 1.1 · 2.2	3.1 · 1.8 · 1.9	91 · 83 · 38	10.1 · 10.3 · 5.7	two passes: the whole cost sits in the second pass that live cannot run (`KeyError: 1`); safety-zone pays MobileSAM + a second vehicle tracker rows only; their cost is what they drag in: the vehicle head +5.8 ms, belt-loader tracking +15.7 ms
the 14 pixel-free modules	0.1–1.5	0.6–0.9	0.75	7–20	1.2–4.9	

Twenty modules together on one camera: 56 ms of 125 (GM 22.4 + tracker 26.6 + hand-off 7.3), 4.2 cores on average with 100 % bursts of 8, 18 GB RAM, GPU 24.5 % busy — the shared part is ≈85 % of the frame path and the modules' own work is the small part. In batch the peak stage is download/upload: handrails-on-gse 30.7 + vests 28.2 + safety-handrails 8.4 + gse-chocks 6.1 + aircraft-chocks 5.7 + 3-stop 3.4 + beltloader-chocks 2.7 + bl_rear_cone 2.4 + cargo 1.4 = **89 ms per recorded frame**, 3–4× over the budget; the walk-arounds add 26.1 (pre-departure) and 6.8 (post-arrival) and stay post.

3 · Dependencies between modules: what they share

- Inputs (architecture review, 29 tasks × 32 concepts)**: camera-obstacle status E07 is read by 23 tasks, aircraft layout E03 by 13, arrival context E01 by 11, belt-loader service status E09 by 10, departure context E02 by 7, chocks/cones detections E04 by 6, worker tracks E11 by 5, chock analysis E14 by 5 (the three chock verdicts, gse-chocks, beltloader-chocks), workers-near-equipment E13 by 4, pushback attachment E10 by 3 (main-gear and nose-wheel chocks, pin verification), ground areas E19 by 3 (FOD corridor, safety zone, pushback path — three different polygons), handrail state E16 by 2 (the same EfficientNet classifier), walk-around evidence E18 by 2; one proposed reuse: the learned pin-installation actions (E24, steering) for pin verification, which today uses posture heuristics.
- GM classes**: cone feeds 22 checks, person 17, beltloader 9, back_door 8, front_door 7, chock 5, pushback 4; the derived rows obstacle / side_obstacle (exact copies of boxes of classes 12/3/31/15, synthesised in GM's second run from the front-wheel geometry) are read by 12 modules as if they were detections; only two modules read confidence at all. **Tracker**: 24 of 27 modules derive the stage themselves and 19 read private optical-flow fields (`\_p0`, `\_st`, `\_of\_dots\_lifetime`, `\_status`), so every tracker version change reaches them; the belt loader is the expensive dependency — 9 checks need it, and it costs +15.7 ms of tracking for the first of them.
- Model sharing**: HRNet-W48 ×4 (identical file; one instance per host is EXACT, shared results are not: windows barely overlap and the preprocessing differs — full-frame CLAEH vs YCrCb on the box); EfficientNet-B0 ×6 (handrail ×2 same weights, difference classifier ×2 different weights, camera classifier in GM, hair policy ×2 — sharing beyond the handrail pair is SEMANTIC); MobileSAM ×4 (GM main aircraft, tracker masks, safety-zone, walk-around TDV); five different pose stacks for ten modules (HRNet, SimCC ResNet-50, RTMPose-I, DWPose-I, MoveNet, YOLOv8-pose); safety vests and hair policy both run a person crop — the only pair free of the aircraft and the stage. **Derived tasks**: proper belt-loader approach = 3-stop + hand-signals; aircraft-chocks gives three verdicts; seat belts out of scope.

4 · Where logic is duplicated

- Arrival stage**: the same block (`movement\_anomaly\_custom` from `p0/\_prev\_p0/\_st` → `normal\_moving\_counter > 24` → `have\_arrival\_stage`, pre-arrival = `arrival\_frame > 60`-fps) is copied verbatim in **9 modules** (aircraft-chocks, cargo doors, chocks-and-cones, crew present, FOD walk, lead marshaller, huddle, safety-zone twice, steering); six more carry their own T_arr / T_dep rules (cones-placed 30-fps moving; pushback wing walkers STOPPED for 60 s; wing walkers stop/depart thresholds with reset; walk-arounds first STOPPED and 3-fps or immediate MOVING; pushback-pathway from the airplane status; hand-signals from `arrival\_frame`) — **five definitions of departure** in the code base.
- pushback_attached**: identical rule in aircraft-chocks and pin-verification (IoU > 0.8 with the median pushback box of pass 1 for > 5 s / > 10 s, nose-geometry fallback) — two copies, both the reason those modules need a second pass; pushback wing walkers uses a third (wheel–pushback distance < 315 px, display only). **Belt loader at the door / active**: five variants — 3-stop (relative intersection with the aircraft > 0.2 + stop counters), beltloader-chocks and bl_rear_cone (`bl\_type` == needed + stopped), handrails-on-gse (`bl\_type` + `is\_stopped`), safety-handrails (+ rear-wheel filter + identity swap across track ids), hand-signals (`\_status` + workers ROI). **Aircraft type** jet/airplane recomputed from wing/engine geometry in beltloader-chocks and hand-signals although GM publishes it.
- Pixels and buffers**: every frame read letterboxes to 1280×768 and makes a CHW/RGB copy that nothing uses (`cv\_common/utis/datasets.py`), paid by all pixel modules; full-frame CLAEH copies in three HRNet modules; the 200-frame buffer of steering and the 96-frame buffer of lead-marshaller (6.2 MB per frame); safety-handrails converts 8 frames to use 1. **Own trackers inside modules** behind the shared one: safety-zone (norfair trucks + cv_common Vehicle with pixel optical flow), huddle (norfair cones), seat-belts (norfair persons). **Infrastructure**: 10 different cv_common pins while `db\_worker/ML\_worker.py` is byte-identical in all 19 checkouts; two checkouts import helpers absent at their pin (PlaneArrivalDetector; departure / presence helpers); three DeepSORT id counters collide across classes and Kalman boxes are cast to int, so tracker boxes ≠ GM boxes; rows are joined by position, not frame id.

5 · Module map by stage (WINDOW the module is active in · decision type · inputs · live ms per frame) and the batch load per stage

Stage	Modules: window, decision (INSTANT one detection → Fail · EVENT verdict at the event · DEADLINE Fail if not seen by T · LOOKBACK window before T · WHOLE the whole event), inputs, live ms	batch ms per recorded frame, sum
PRE_ARR · start → T_arr	chocks-and-cones (LOOKBACK, rows, 0.1) · crew-present (LOOKBACK, rows, 0.5) · FOD walk (LOOKBACK, pixels, MoveNet, 1.3) · huddle (LOOKBACK, pixels, no model, 0.9) · lead-marshaller (LOOKBACK, pixels, HRNet, 2.5) · safety-zone (T_arr – 2 min → T_arr, two passes, MobileSAM + own tracker, RETHINK)	51.7 (vests 28.2 of it)
ARR_POST · T_arr → ...	cones-placed (DEADLINE T_arr + limit, rows, 0.6) · steering (DEADLINE T_arr + 240 s, pixels, HRNet + InceptionTime, 0.7) · hand-signals (T_arr → GSE at the door, two passes, HRNet, RETHINK) · nose gear chocks (within 30 s of the stop; aircraft-chocks, two passes, EfficientNet difference) · post-arrival walk-around (WHOLE, POST, own environment)	48.2 (+6.8 post)
DOWNLOAD · BL appears → BL leaves	3-stop (BL appeared → stopped at the door, intersection > 0.2; EVENT, rows, 0.6) · beltloader-chocks (BL at door → drove away, EVENT, rows, 0.3) · bl_rear_cone (EVENT, rows, 0.4) · cargo doors (DEADLINE end of stage / 480 frames, rows, 0.9) · gse-chocks (GSE stopped → drove away, EVENT, pixels, RTMPose + EfficientNet, 3.3, session end) · handrails-on-gse (BL at door → worker off the BL, EVENT, pixels, four networks, 12.5, session end) · safety-handrails (BL at door → drove away, EVENT, pixels, EfficientNet, 2.4)	89.0 — the peak, 3–4× the budget
PRE_DEP · BL leave → T_dep	conditioned-air (LOOKBACK T_dep – 10 min, pixels, YOLO-seg every 16th frame, 1.3) · cones-removed (EVENT BL leave → T_dep, rows, 1.5) · pin-verification (BL leave → pushback attached, two passes, SimCC pose, PATCH) · main-gear and nose-wheel chock removal (after pushback attached; aircraft-chocks, two passes) · pre-departure walk-around (WHOLE, POST, own environment)	45.1 (+26.1 post)
DEP · pushback → T_dep + 60 s	pushback wing walkers (pushback attached → started moving, EVENT, rows, 0.3) · pushback-pathway (T_dep – 45 s → T_dep, INSTANT, rows, 0.5) · wing-walkers (T_dep – preparation → T_dep + 60 s, LOOKBACK, pixels, HRNet, 5.1)	42.9
ALL · start → end	safety-vests (INSTANT: an unzipped vest → Fail, but the client verdict only at session end; pixels, two Swin-T + ResNet-18, 18.8, the only check free of the aircraft and the stage) · hair-policy (WSL, CPU, not hosted)	28.2 all session

What this means for the shared layer: one stage detector replaces 9 + 6 copies of the stage and 2 of pushback_attached; one HRNet instance per host and the frame-reader fix are EXACT wins; one pose stack and shared segmentation masks are SEMANTIC and need the verdict gate; the belt-loader state (five variants) belongs in the tracker's published records, not in the modules.